Palm OS Programming Recipes

Exchange Manager: Sending Text to Local PIM Applications in Palm OS&153; 4.0 or Later

Starting with Palm OS® version 4.0, the Exchange Manager works in conjunction with an exchange library. Each exchange library is transport-dependent and performs the actual communication with the remote device. Currently, four exchange libraries are supported by Palm OS 4.0, including the IR library(IrDA), the local exchange library, the SMS library, and the Bluetooth library. In Palm OS 3.x, the only exchange library available is the IR Library, and it is not extensible.

If you want to use the Exchange Manager to exchange data with other applications on the local device in Palm OS 4.0, you can use the Local Exchange Library. In Palm OS 3.x, you need to set the `localMode` flag to `1` to use the local device mode of the IR Library. Please refer to other recipes for details.

Note: The programming technique you used for the Exchange Manager in Palm OS 3.x will also work under Palm OS 4.0. See the recipe, "Sending Text to Local PIM Applications in Palm OSª 3.0 or Later," for an example.

The following recipe shows you how to send a message to the local Memopad application using Palm OS 3.0 or later. It will send a string buffer, with the value "test," as a .txt data type to Memopad. You should be focused on how to set up the data structure `ExgSocket`, which is the core of the Exchange Manager.

Follow these steps:

In Constructor:

- Add a menu item for sending or any for any other UI components that are applicable to your application.

Open the source file (`*.c`) and find the event loop handling function that will handle the "Send" menu item.

- Make sure that `ExgMgr.h` is included as a header file.
- Create and initialize an `ExgSocketType` data structure with information about which library to use and the data to be sent. Make sure to initialize the description, type, target, and/or name.
- Set `name` as a local URL to tell the Exchange Manager to send to local device, for example, `"?_local:Sample.txt"`.
- Set `name` to end with .txt. This is the file type that Memopad is registered to receive.
- Call the `ExgPut` function to establish the connection with the exchange library.
- Call the `ExgSend` function to send the actual data in a loop (if the sending buffer is big).
- Call the `ExgDisconnect` function to end the connection.

These code segments are examples of the above procedure:

```
        static void SendToLocalMemopad()
        {
            ExgSocketType exgSocket;
            Err err = 0;
            // message that will be sent to memopad
            Char        *textBuf = "test";
            UInt32 size = StrLen(textBuf) + 1;

            // it's important to initialize structure to zeros
            MemSet(&exgSocket,sizeof(exgSocket),0);
            exgSocket.description = "Testing message from
                ExgMgr40LocalPIM";

            // Tips for using the new Exchange Manager in
            // Palm OS 4.0:
            // If more than one exchange library is
            // registered
            // for the designated schemes, the Exchange
            // Manager has the user select the exchange
            // library by displaying the Send With dialog
            // (controlled by the "?").
            // The "name" field needs to have ":" at the
            // end of schemes.
            // "_local" is the value of predefined scheme constant
            // "exgLocalPrefix."
            exgSocket.name = "?_local:Sample.txt";

            // An optional line of code to initialize "type" for socket.
            // This code only works in Palm OS 4.0.
            // Palm OS 4.0 will check this field for the target
            // If this field is not set (it is zero), it will look
            // for the extension in
    // the "name" field. So the next line of code is
    // optional if ".txt" is
    // given in the "name" field.
            exgSocket.type = ".txt";

            /* The old way of sending in Palm OS 3.x still
            // works on Palm OS 4.0.
            // will comm. with memopad app
            exgSocket.name = "Sample.txt";
            // set to comm. with local machine(PIM) only
            exgSocket.localMode = 1;
    */

            err = ExgPut(&exgSocket);
            if (err == 0) {
          ExgSend(&exgSocket,textBuf,size,&err);
          ExgDisconnect(&exgSocket,err);
    }
        }
```

As you've seen in the code above, in Palm OS 4.0 and higher, you will use "next" assignment in your program to tell the Exchange Manager to send to local devices (URL assignment):

```
exgSocket.name = "?_local:BeamSample.txt";
```

"`_local`" is the string value of the predefined scheme constant `exgLocalPrefix`. The

question mark ("?") is based on the next consideration and is optional. If more than one exchange library is registered for the provided schemes, the Exchange Manager has the user select the exchange library by displaying the Send With dialog. Please check the SDK documentation for details.

Sample program downloads:

- Windows
- Mac